HDOC Day-13

1. **Write a menu driven C Program using switch-case to input a positive integer and perform one of the following operations based on the user's choice.**

- **Count the number of even digits, and odd digits. E.g- Number:12045,Even_dig= 3, Odd_digit = 2**
- **Find the sum of its even digits and the sum of its odd digits separately. E.g- Number:12345, Sum_even = 6, Sum_odd = 9**

2. Algorithm

Step 1:

- Start the program

Step 2:

- Input a positive integer n

Step 3:

- Display the menu:

Choice 1: Count even and odd digits

Choice 2: Find sum of even and odd digits

Step 4:

- Input the user's choice.

Step 5:

- Use switch-case according to the choice.

- ### For Choice 1:

Set even_count = 0 and odd_count = 0.

Extract the last digit using digit = n % 10.

Check whether digit % 2 == 0.

If true, increment even_count; otherwise increment odd_count.

Remove the last digit using n = n / 10.

Repeat until n becomes 0.

Display the number of even and odd digits.

For Choice 2:

Set sum_even = 0 and sum_odd = 0.

Extract each digit using digit = n % 10.

If the digit is even, add it to sum_even.

Otherwise, add it to sum_odd.

Remove the last digit using n = n / 10.

Repeat until n becomes 0.

Display both sums.

Step 6:

- If the choice is invalid, display Invalid Choice.

Step 7:

- Stop the program

• 3. C Program:

```
#include <stdio.h>


int main()
{
  int n, choice, digit;
  int even_count = 0, odd_count = 0;
  int sum_even = 0, sum_odd = 0;
  printf("Enter a positive integer: ");
  scanf("%d", &n);
```

```
printf("\nMENU\n");
printf("1. Count even and odd digits\n");
printf("2. Find sum of even and odd digits\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
  case 1:
    while(n > 0)
    {
      digit = n % 10;

      if(digit % 2 == 0)
     even_count++;
      else
        odd_count++;

      n = n / 10;
    }

    printf("Number of even digits = %d\n", even_count);
    printf("Number of odd digits = %d\n", odd_count);
    break;

  case 2:
    while(n > 0)
```

```
        {
            digit = n % 10;


            if(digit % 2 == 0)
                sum_even += digit;
else
                sum_odd += digit;


           n = n / 10;
        }


        printf("Sum of even digits = %d\n", sum_even);
         printf("Sum of odd digits = %d\n", sum_odd);
         break;


       default:
         printf("Invalid choice!\n");
    }


    return 0;
}
```

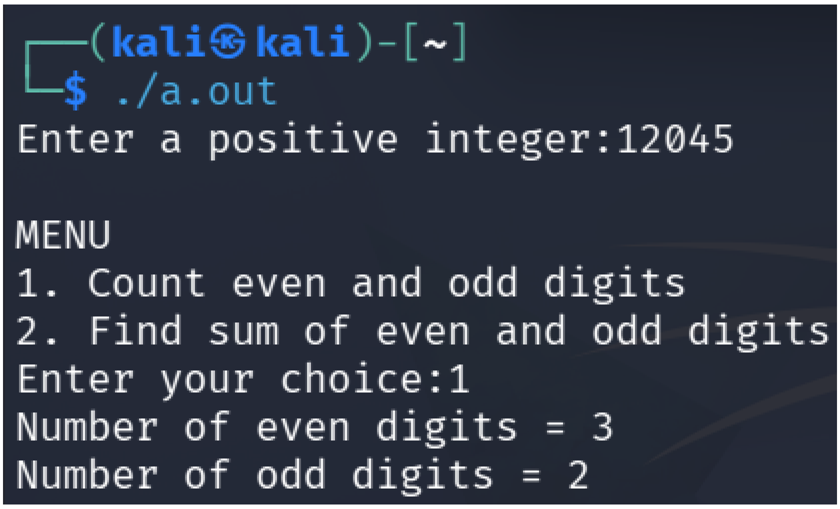
Test Case Table:

Test Case	Input Number	Choice	Expected Output
1	12045	1	Even digits= 3 , Odd digits=2
2	12345	2	Sum of even digits=6 , Sum of odd digits= 9
3	2468	1	Even digits=4, Odd digits=0

4	13579	2	Sum of even digits=0, Sum of odd digits= 25
5	123456	1	Even digits=3, Odd digits=3
6	123456	2	Sum of even digits=12 , Sum of odd digits= 9
7	12345	3	Invalid Choice

For choice 1:

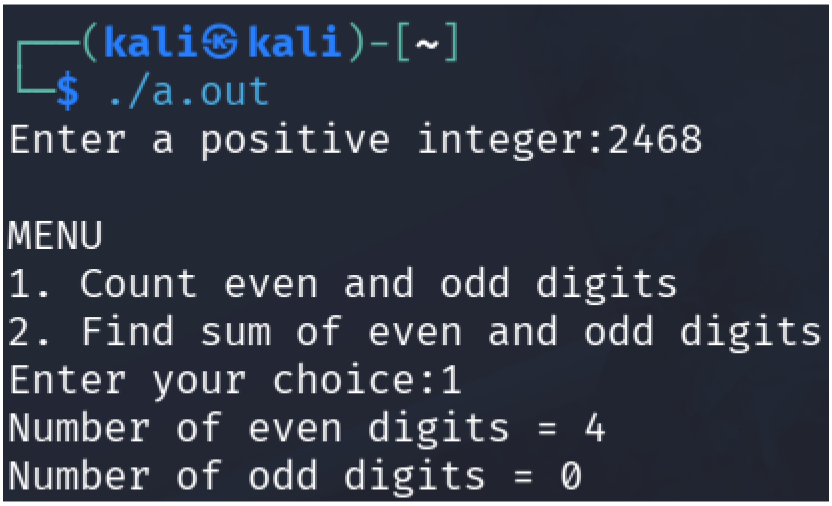
Test Case1 :



```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:12045

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:1
Number of even digits = 3
Number of odd digits = 2
```

Test Case 2:



```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:2468

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:1
Number of even digits = 4
Number of odd digits = 0
```

Test Case 3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:123456

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:1
Number of even digits = 3
Number of odd digits = 3
```

For Choice 2:

Test Case 1:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:2
Sum of even digits = 6
Sum of odd digits = 9
```

Test Case 2:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:13579

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:2
Sum of even digits = 0
Sum of odd digits = 25
```

Test Case 3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:123456

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:2
Sum of even digits = 12
Sum of odd digits = 9
```

For Choice 3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a positive integer:12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice:3
Invalid choice!
```